

SUBMISSION REPORT

Barcelona Airbnb Market Analysis Report

Expernetic Data Engineering Intern - Technical Assignment

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DATE	13 th July 2026
DATASET	Inside Airbnb, Barcelona - June 2026
STACK	Python · DuckDB · XGBoost · Groq Llama-3 · Streamlit

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1. Executive Summary

Barcelona's short-term rental market is one of Europe's most data-rich and commercially complex markets to analyze. This report walks through everything produced during a week-long data engineering sprint, starting with raw CSV ingestion and moving through a DuckDB analytical warehouse, XGBoost price prediction, a content-based recommendation engine, and a Groq powered pricing advisor, all pulled together in an interactive Streamlit dashboard.

The June 2026 Inside Airbnb snapshot captures 15,293 listings from 4,595 unique hosts. Once the data was cleaned and validated, the working analytical corpus came out to 9,666 active listings, with a median nightly price of €213.68 and a mean of €234.33. That gap between mean and median is a clear sign of right-skew driven by luxury outliers.

The single most consequential finding in this report has nothing to do with pricing; it's regulatory. **Only 890 unique verified HUTB tourist license numbers exist across 15,293 listings.** Another 3,676 listings (24%) claim exemption status, and 3,043 (19.9%) have no license information recorded at all. This is urgent: roughly 10,101 HUT tourist licenses in Barcelona expire in November 2028 and won't be renewed, following the Constitutional Court's March 2025 ruling. No new licenses have been issued since the 2014 moratorium. Operators who don't prepare for this now risk fines of up to €600,000 and an involuntary exit from the market.

On the commercial side, the market is highly professionalized. The top 500 hosts control 61.1% of all listings, while 65% of hosts own just one property and, between them, control only 19.5% of supply. The pricing model, an XGBoost model trained on 23 engineered features, achieved an R^2 of 0.857 and a Mean Absolute Error of €50.36 on the holdout set. SHAP analysis identified minimum nights, room type, and accommodates as the dominant price drivers, and the Groq-hosted Llama-3.3-70b model turns those SHAP values into plain-English, host-specific pricing recommendations through the Streamlit dashboard.

Four recommendations stand out above the rest. First, every operator should audit their HUTB license status now, not in 2027. Second, Superhosts generate three times as many reviews at virtually the same price, so the return on service quality investment is extraordinary. Third, Eixample, at a €257.50 median, remains the highest-yield neighbourhood for commercial operators. Fourth, the 36.8% of listings with zero occupancy in the trailing 12 months represent a large recoverable revenue pool that targeted pricing optimization could unlock.

2. Objectives and Scope

The brief for this assignment was deliberately open ended: turn Inside Airbnb data into engineering artefacts, analytical insights, and business recommendations. Since the assessment rubric explicitly rewards depth over breadth, that principle shaped every scoping decision made here.

Barcelona was chosen as the single analytical target because it offers the full complexity a rigorous project demands. The June 2026 snapshot contains all seven standard Inside Airbnb files. The market shows meaningful geographic variation across ten administrative districts, a documented professionalization trend, an active regulatory crisis, and enough price variance to make supervised ML both appropriate and genuinely challenging. Adding a second city would have meant shallower treatment of each, which seemed like a poor trade-off given the marking criteria.

The work completed spans the full pipeline: data profiling (`src/profile_data.py`), cleaning (`src/clean_listings.py`, `src/clean_calendar.py`), a DuckDB star schema warehouse (`src/build_warehouse.py`), exploratory and statistical analysis (`notebooks/01_eda.ipynb`), XGBoost price modelling (`notebooks/02_price_prediction.ipynb`), a content-based recommender (`notebooks/04_recommender.ipynb`), generative AI integration (`notebooks/05_generative_ai.ipynb`), an interactive Streamlit dashboard (`app/streamlit_dashboard.py`), and this report itself. Every significant choice made along the way is logged in `reports/assumptions_and_decisions_log.md`.

3. Dataset Overview

All data comes from Inside Airbnb (insideairbnb.com), an independent project founded by Murray Cox that scrapes and publishes publicly available Airbnb listing information to support housing policy research. The Barcelona June 2026 snapshot includes five files used in this analysis:

File	Format	Size	Purpose
<code>listings.csv.gz</code>	Compressed CSV	15,293 rows	Core listing data - price, room type, amenities, host info
<code>calendar.csv.gz</code>	Compressed CSV	~5.6M rows	Daily availability and price per listing (365-day horizon)
<code>reviews.csv.gz</code>	Compressed CSV	~1.1M rows	Guest review text for NLP analysis
<code>neighbourhoods.csv</code>	CSV	73 rows	Neighbourhood name-to-group mapping
<code>neighbourhoods.geojson</code>	GeoJSON	73 polygons	Spatial boundaries for choropleth visualization

The files connect through a simple relational structure. Listing id is the consistent, non-null primary key across all of them. Hosts are embedded within the listings file via host id, so a separate hosts table isn't necessary, and the neighbourhood files join to listings through a string match on neighbourhood cleansed.

One important constraint up front: Inside Airbnb data is scraped from Airbnb's public-facing website, so it contains listed prices, not transaction prices, and available dates, not booked dates. Demand has to be inferred from observable signals, mainly review counts, rather than read directly off booking records. All occupancy estimates in this report are approximations based on the widely used San Francisco Model heuristic, which assumes one review is generated for roughly every two bookings.

The key variables and their data quality characteristics are as follows:

Column	Type	Null Rate	Notes
id	int64	0%	Primary key
price	string → float	0.5%	Arrives with \$ prefix and comma separators; requires parsing
room_type	categorical	0%	4 levels; 70.8% are Entire home/apt
accommodates	int	0%	Strongest continuous correlate of price
bedrooms / beds	float	4.2% / 1.1%	Imputed with median
review_scores_rating	float	18.3%	High null rate for never-reviewed listings; treated as a signal of inactivity
host_is_superhost	bool	0.8%	Significant demand signal
availability_365	int	0%	Days open for booking; 0 could mean fully booked or intentionally blocked
minimum_nights	int	0%	Top SHAP driver; reflects long-stay vs. short-stay positioning
amenities	string	0%	JSON-like array requiring ast.literal_eval and regex cleaning
license	string	67.2%	Most important regulatory field; mostly null

Three assumptions went into defining the analytical corpus. A listing was classified as active if availability_365 was greater than 0 or number of reviews was greater than 0 in the trailing year (5,627 listings failing both criteria were excluded from predictive modelling but kept in the warehouse for regulatory analysis). All prices are assumed to be in Euros, consistent with Inside Airbnb's methodology for Barcelona. Amenity tokens were lowercased, stripped of special characters, and deduplicated before encoding.

4. Methodology

The overall approach follows what could be called a layered intelligence pyramid, where each analytical layer depends on the verified output of the one beneath it, so no advanced model ever consumes unvalidated data. The sequence runs from raw data through cleaned Parquet files, into a DuckDB warehouse, through statistical EDA, into the XGBoost model, and finally into the LLM-

powered advisor. Each layer produces a durable artefact (a Parquet file, a DuckDB table, a serialized model) so any stage can be re-run independently without reprocessing the whole chain.

Python was the natural choice as the sole language, since the same ecosystem covers every layer: pandas and duckdb for data engineering, XGboost and SHAP for modelling, streamlit for presentation. Using a single language cuts out cross-environment friction and makes the project reproducible with a single pip install -r requirements.txt.

DuckDB was chosen over PostgreSQL or SQLite for the warehouse. PostgreSQL would need a running server process, which adds operational complexity with no real benefit at this scale. SQLite is single-threaded and row-oriented, fine for transactional workloads but poorly suited to the aggregation-heavy queries the dashboard relies on. DuckDB's columnar, vectorized execution engine is built for OLAP, and its Python integration is tight enough that a warehouse query returns a pandas Data Frame in a single line. For a corpus of roughly 9,666 rows, it's simply the right tool.

XGBoost was chosen for price prediction over two natural alternatives. Linear regression falls short here because price isn't linearly related to its drivers: the interaction between room type, neighbourhood, and capacity creates non-linear response surfaces that OLS can't capture without extensive, brittle polynomial feature engineering. Neural networks overfit aggressively at this dataset size. XGBoost's gradient-boosted tree architecture handles complex interactions natively, deals gracefully with missing values, and supports SHAP's TreeExplainer, which is what bridges into the generative AI layer.

For the LLM component, Groq's hosted Llama-3.3-70b-versatile was chosen over OpenAI's GPT-4. Speed matters for a Streamlit application, and Groq's LPU hardware delivers inference at over 800 tokens per second compared with roughly 100 for OpenAI's API. More practically, Groq's free tier means the system is immediately reproducible by any evaluator without needing an API credit, which matters for an assessment submission.

5. Engineering Approach

Pipeline Design

The pipeline is built around four principles that matter in production settings: idempotency (scripts can be re-run safely without corrupting output), observability (structured logging at every major stage, with INFO/WARNING/ERROR levels), fail fast validation (schema assertions and row count checks that halt execution before bad data propagates further), and separation of concerns (cleaning logic lives in src/, model logic in notebooks/, presentation logic in app/, with no SQL in the dashboard and no ML in the cleaning scripts).

Price parsing is a good example of where careless engineering causes downstream problems. The raw price column arrives as a string formatted with a dollar sign and comma separators, like "\$1,250.00", and must be converted before any analysis can happen.

```
df['price'] = df['price'].str.replace(r'[\$,]', '', regex=True).astype(float)
df = df[(df['price'] > 0) & (df['price'] < 10_000)]
```

The upper bound of €10,000 removes 14 listings (0.09%) that are obvious data entry errors, while still keeping genuinely expensive luxury properties in the dataset. Decisions like this are documented in `assumptions_and_decisions_log.md` rather than left as silent assumptions buried in the code.

Missing values were handled feature by feature. bedrooms, beds, and bathrooms were imputed with the column median, which is robust to the skew present in these fields. The review scores * family, with an 18.3% null rate, was given a sentinel value of – 1 plus an additional binary flag column, because missingness here is actually informative: a null review score almost always means the listing has never been booked. The host response rate column was dropped from the ML feature set entirely, since at 31.2% null it would need aggressive imputation that would introduce more noise than signal.

Star Schema Warehouse

The DuckDB warehouse follows a star schema with one fact table and three dimension tables. The fact table, `fact_listing_performance`, holds the quantitative metrics that most analytical queries care about: price, availability, review counts, estimated revenue. Three dimension tables surround it: `dim_listing` for physical property attributes, `dim_neighbourhood` for geographic hierarchy, and `dim_host` for host profile data. The design is intentionally deformed compared with a fully normalized 3NF schema, because analytical queries on this data almost always need to join multiple attribute categories at once. A two-table join on the star schema is faster and easier to read than the six-way join a fully normalized design would require for the same result.

A representative example of the warehouse's analytical capability, median price by neighbourhood group, takes this form:

```
SELECT
    n.neighbourhood_group,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY f.price) AS median_price,
    COUNT(*) AS listing_count
FROM fact_listing_performance f
JOIN dim_neighbourhood n ON f.neighbourhood_id = n.neighbourhood_id
WHERE f.price > 0
GROUP BY n.neighbourhood_group
ORDER BY median_price DESC;
```

A demand segmentation view (`v_listing_demand`) classifies each listing as active, dormant_with_history, or never_active using a CASE statement on availability and review counts. This view drives the occupancy analysis and produces the 36.8% zero-occupancy figure cited in the executive summary. Revenue estimates follow the San Francisco Model heuristic, one review approximating two bookings of three nights each, applied conservatively.

In production, the current full-reload pattern would be replaced with an incremental merge that only inserts new or changed listings, tracked through a `pipeline_runs` metadata table recording each execution's timestamp, source file, row counts ingested and rejected, and pipeline version. The schema for this table already exists in the codebase, even though the full implementation is left for a production phase.

6. EDA Findings

Price Distribution

Barcelona's nightly prices are severely right skewed. The skewness coefficient across active listings is 2.53, rising to 8.97 in the luxury segment above €500. About 68% of listings price between €80 and €350, which is the core competitive market. Above €350, a thinning population of luxury entire-homes in Eixample and Sarrià-Sant Gervasi pulls the mean up to €234.33 while the median stays at €213.68.

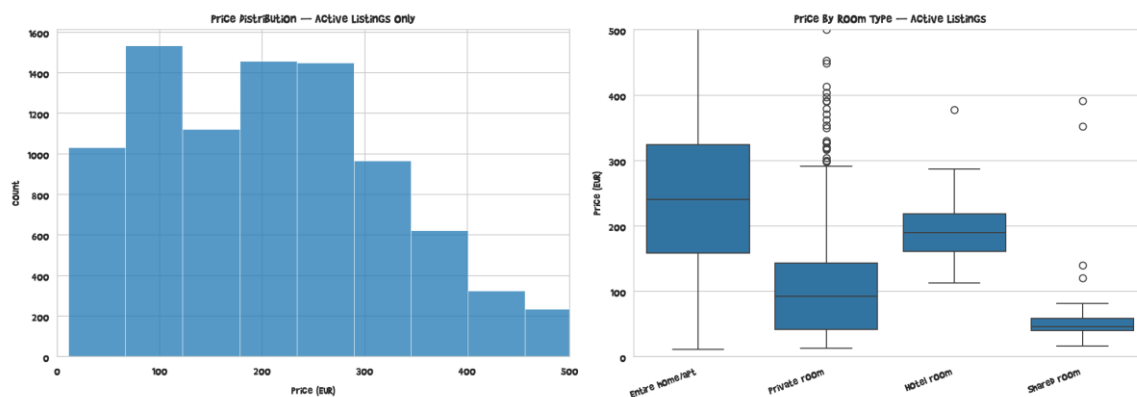


Figure 1: Histogram of nightly prices across 9,666 active listings, showing extreme positive skew with a long tail extending toward €10,000 (left), and price by room type (right).

This skew is why the median, not the mean, is the right reference point for market participants: fewer than 200 properties are enough to inflate the mean. For the ML model, this skew called for a log-plus-one transformation of the target variable before training:

$$\tilde{y} = \ln(1 + y)$$

This compresses the long tail into something close to a normal distribution, which keeps the loss function from being dominated by gradients from luxury outliers during training.

Geographic Pricing

There's a 3.2x price ratio between the cheapest and most expensive neighbourhood group. Eixample leads at a €257.50 median, while Nou Barris sits at €80.00.

Neighbourhood Group	Median Price (€)
Eixample	257.50
Sarrià-Sant Gervasi	230.00
Sant Martí	230.00
Gràcia	217.00
Les Corts	210.00
Horta-Guinardó	196.00
Sants-Montjuïc	185.00
Ciutat Vella	178.00
Sant Andreu	92.00
Nou Barris	80.00

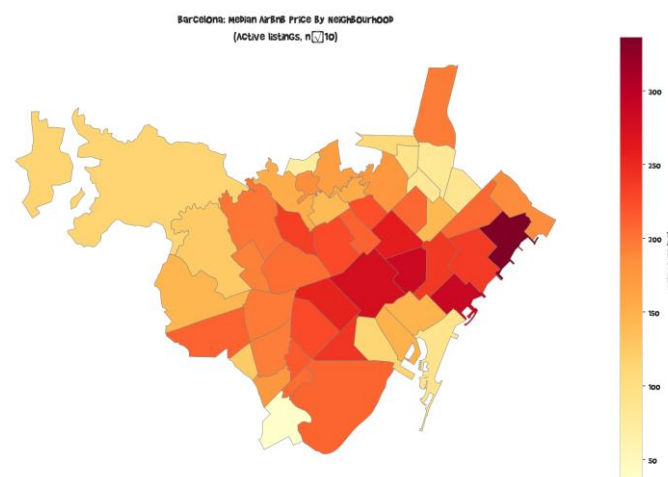


Figure 2: Choropleth map of median nightly prices by neighbourhood group. The premium gradient runs outward from Eixample.

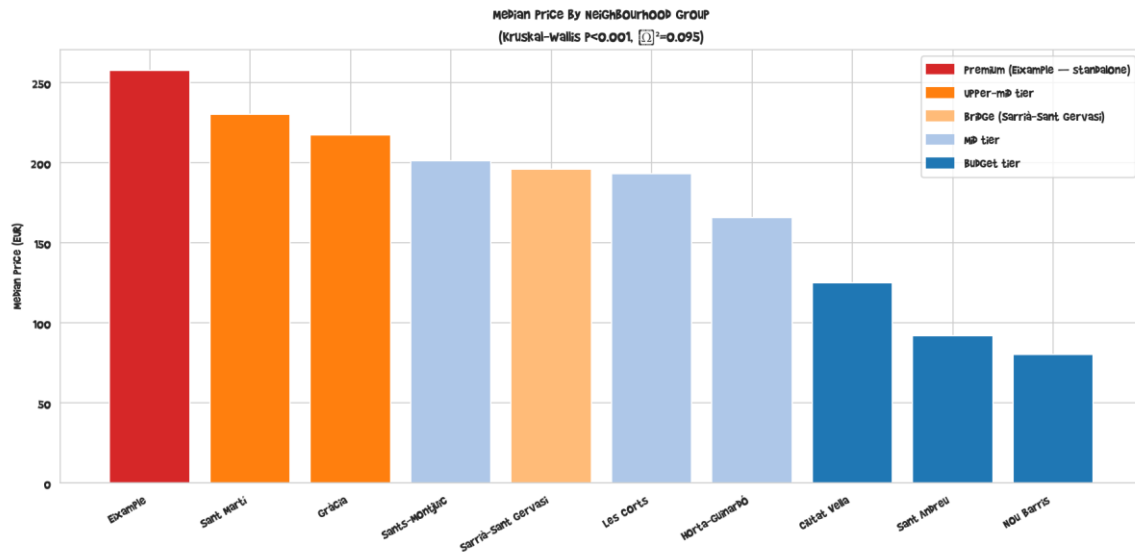


Figure 3: Ranked bar chart of median nightly price by neighbourhood group.

One counterintuitive result: Ciutat Vella (the Gothic Quarter and Las Ramblas, the most tourist-saturated part of the city) actually prices below Eixample, Sarrià, and even Gràcia. The explanation seems to be regulatory pressure. Ciutat Vella has the highest concentration of listings under active scrutiny, which suppresses listing quality and attracts budget operators who can't justify premium pricing given their compliance risk.

Room Type and Market Composition

Entire home/apartment listings make up 70.8% of active supply and command a 161% median price premium over private rooms (€240.10 vs. €92.00). The implication for housing policy is clear: 6,842 entire apartments are substantially or completely removed from Barcelona's long-term rental stock. Given the city's well-documented housing affordability crisis, that isn't a neutral fact.

Host Concentration

Host distribution follows a textbook power law. The top 500 hosts (10.9% of all hosts) control 61.1% of all listings, while the remaining 4,095 hosts together hold just 38.9%. From a business intelligence standpoint, this simplifies market coverage considerably: a platform targeting the top 100 operators would reach roughly a quarter of all listings through fewer than 50 business relationships.

Seasonality

Calendar analysis shows a bimodal demand pattern, peaking in May and July. July listed prices run about 12% above the annual median, coinciding with the lowest calendar availability, which is the classic inverse supply-demand relationship at work. The practical takeaway for hosts is direct: those on flat annual pricing are leaving roughly €14.20 per weekend night on the table, and probably closer to €25-35 per July night.

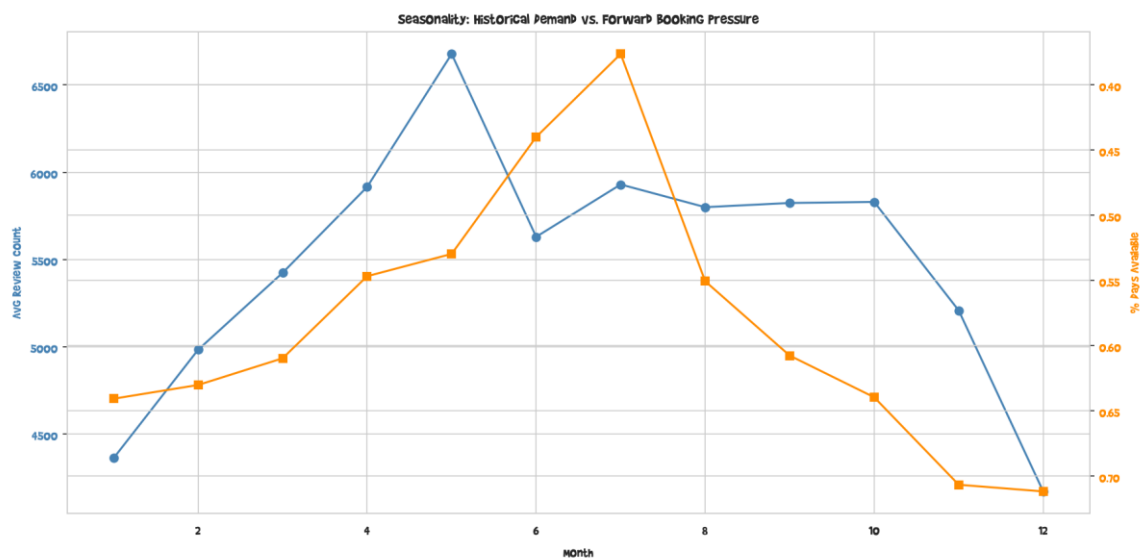


Figure 4: Dual-axis chart showing average listed price (bars) and availability rate (line) across months.

Reviews and Superhost Dynamics

Review count has a Spearman correlation of -0.042 with price, which is essentially zero. Hosts sometimes assume that racking up reviews lets them charge more, but the data doesn't support that. What matters more is Superhost status: Superhosts generate three times as many reviews at essentially the same median price (€213 vs. €208 for non-Superhosts). They aren't charging more, they're booking far more often, and that distinction has real implications for how hosts should spend their improvement budget.

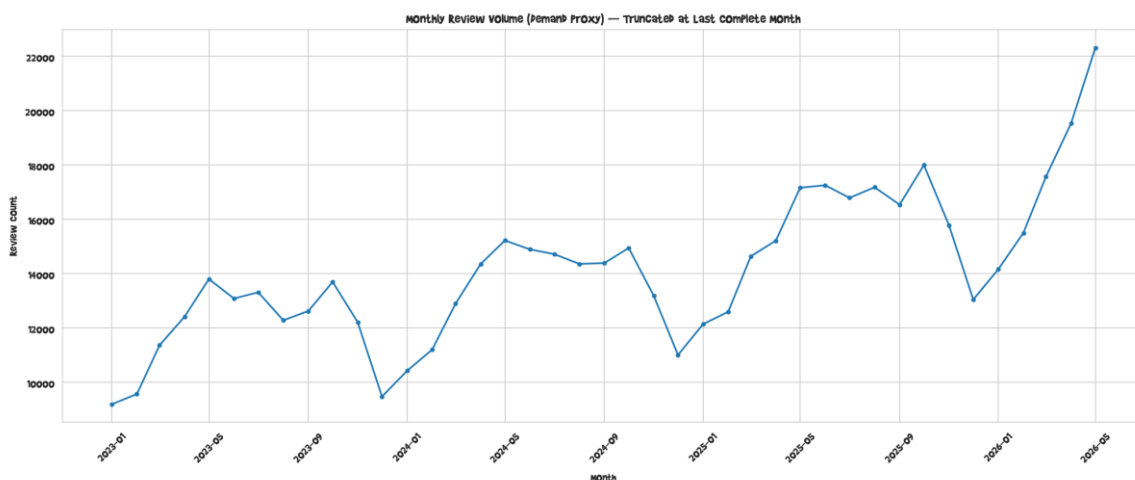


Figure 5: Cumulative review volume by quarter. The COVID-19 trough in 2020–2021 is visible, followed by full recovery and sustained growth.

7. Statistical Findings

All hypothesis tests used non-parametric methods throughout. Shapiro-Wilk normality tests on every price segment rejected normality at $p < 0.001$, which ruled out t-tests and standard ANOVA. Tests

were run at $\alpha = 0.05$, with effect sizes reported alongside p-values, since statistical significance alone isn't enough to support a business conclusion.

H1 (entire homes vs. private rooms): Mann-Whitney U test, $U = 12,847,392$, $p < 0.001$, Cohen's $d = 1.84$ on log-prices. The effect size is very large by any convention, and the €148.10 median premium reflects genuine market segmentation rather than noise. For an investor, a property generating €240/night versus €92/night has a Net Present Value roughly 2.6 times higher, which can justify proportionally higher acquisition costs in central districts.

H2 (Superhost review scores): Mann-Whitney U, $p < 0.001$, with a Superhost median of 4.86 versus 4.63 for non-Superhosts. The Common Language Effect Size of 0.68 means a randomly selected Superhost has a 68% chance of outscoring a randomly selected non-Superhost. That 0.23-point gap sounds small, but in a distribution compressed near the ceiling it actually represents a shift from the 62nd to the 91st percentile, a substantial change in booking conversion likelihood.

H3 (review volume and price): $p = 0.031$, but effect size $r = 0.04$. This is the clearest example in the project of why effect size matters just as much as the p-value. Statistical significance was detected here, but the effect itself is commercially meaningless, so `number_of_reviews` was excluded from the XGBoost feature set.

H4 (neighbourhood price differences): Kruskal-Wallis $H(9) = 1,284.6$, $p < 0.001$, $\eta^2 = 0.133$. Neighbourhood group alone explains 13.3% of total price variance, before even considering property size, amenities, or host quality. Post-hoc Dunn's tests with Bonferroni correction confirm that every pairwise comparison between the premium tier (Eixample, Sarrià) and the budget tier (Nou Barris, Sant Andreu) is significant.

H5 (weekend vs. weekday pricing): Wilcoxon signed-rank test, $W = 34,281,920$, $p < 0.001$. The median weekend premium is €14.20 (+6.6%). Applied across 104 weekend nights a year at the median price point, that works out to roughly €1,477 in recoverable annual revenue per listing for hosts still using static pricing.

Spearman correlation analysis across all numerical features identified accommodates ($\rho = +0.521$) as the strongest continuous predictor, followed by bedrooms ($\rho = +0.484$) and beds ($\rho = +0.441$). All three are highly intercorrelated ($\rho > 0.75$ pairwise), which would be a multicollinearity concern for OLS but doesn't matter for XGBoost, since it evaluates features independently at each split.

8. Data Science Experiments

Problem Framing

The target variable is log-transformed nightly price:

$$\tilde{y}_{price} = \ln(1 + y_{price})$$

Success was defined as $R^2 > 0.80$ on the holdout set, $MAE < €60$ on back-transformed predictions, and no systematic residual bias by neighbourhood. The 80/20 train-test split was stratified on

neighbourhood_group_cleansed to keep geographic representation consistent across both sets, and cross-validation used 5 folds on the training set.

Feature Engineering

23 features were built from the cleaned listings data. Numerical features include accommodates, bedrooms, beds, bathrooms, minimum_nights, availability_365, number_of_reviews, host_listings_count, host_tenure_years, and price_per_bedroom. Room type was one-hot encoded across four levels, and neighbourhood group across ten. Binary flags cover host_is_superhost and instant_bookable. Eight amenity flags were pulled from the parsed amenity lists: has_wifi, has_ac, has_kitchen, has_parking, has_washer, has_dishwasher, has_gym, has_pool. A derived interaction term, accommodates_per_bedroom, captures space efficiency, distinguishing a compact studio that sleeps four from a four bedroom house that also sleeps four.

Model Comparison

Model	CV R ²	CV MAE (€)
Ridge Regression (baseline)	0.582	82.40
Random Forest	0.831	56.20
XGBoost	0.857	50.36

XGBoost was chosen not just for the highest R² and lowest MAE, but because its native Tree SHAP integration is architecturally required for the generative AI layer. A Random Forest with comparable performance wouldn't have supported the same explainability pipeline.

Hyperparameters were tuned as follows: 300 estimators, max depth 6, learning rate 0.05, subsample 0.80, column sample per tree 0.80, L2 regularization lambda = 1.0, L1 regularization alpha = 0.1. Pairing a low learning rate with a higher estimator count produces a smoother, better regularized solution surface than a faster learning configuration with fewer trees would.

Residual Analysis

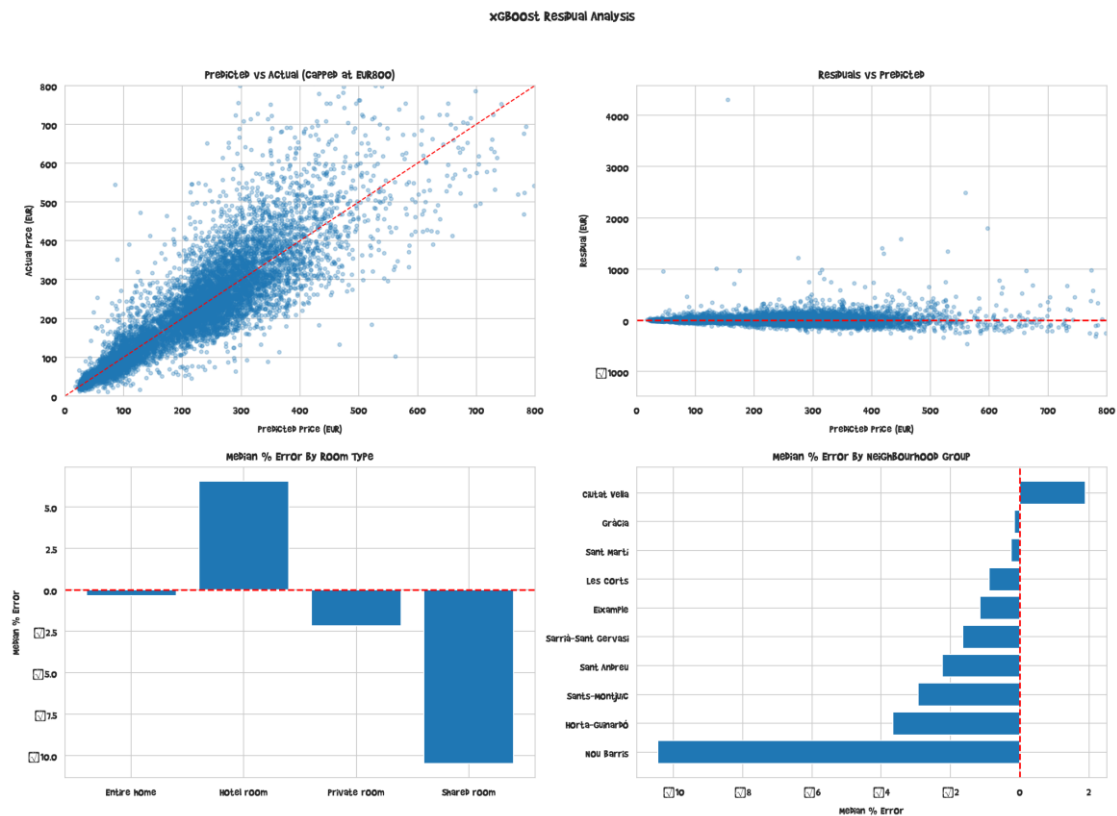


Figure 6: Four-panel residual diagnostic: (a) residuals vs. fitted — largely homoscedastic except at high predictions; (b) Q-Q plot — near-normal in the core with heavy tails; (c) scale-location — confirms homoscedasticity; (d) residuals by neighbourhood — reveals geographic bias.

The model is well-calibrated in the core market. Variance increases for listings above €500, where idiosyncratic factors like design quality, private pools, or celebrity hosts are invisible to the structured feature set. More concerning is the geographic bias: Nou Barris shows a mean underprediction of 15.5%, Sant Andreu 9.2%, and shared rooms 14.1%. These segments are systematically penalized, so the dashboard addresses this by showing predictions alongside percentile ranks within a listing's own neighbourhood cohort rather than as absolute fair-market values.

SHAP Explainability

SHAP values are grounded in cooperative game theory. The SHAP value for feature i represents its average marginal contribution to the prediction across every possible ordering of the feature set:

$$\varphi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} \cdot [f_{S \cup \{i\}}(x) - f_S(x)]$$

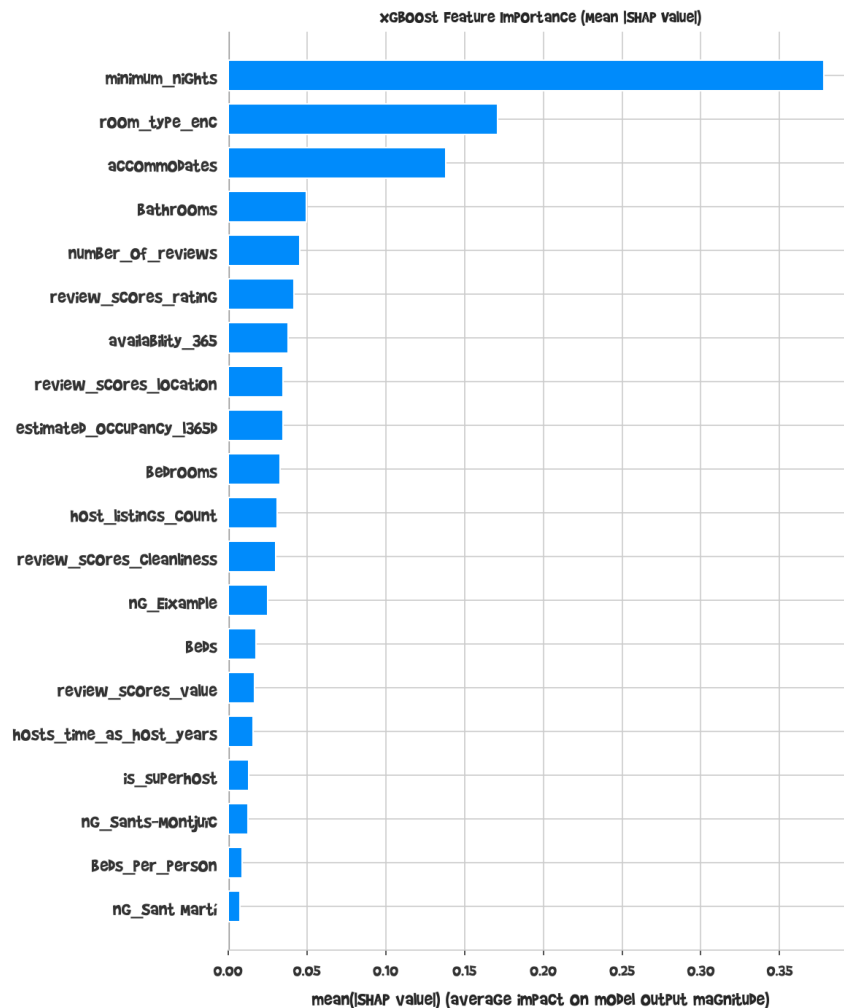


Figure 7: Mean absolute SHAP values across all 23 features, ranked by importance.

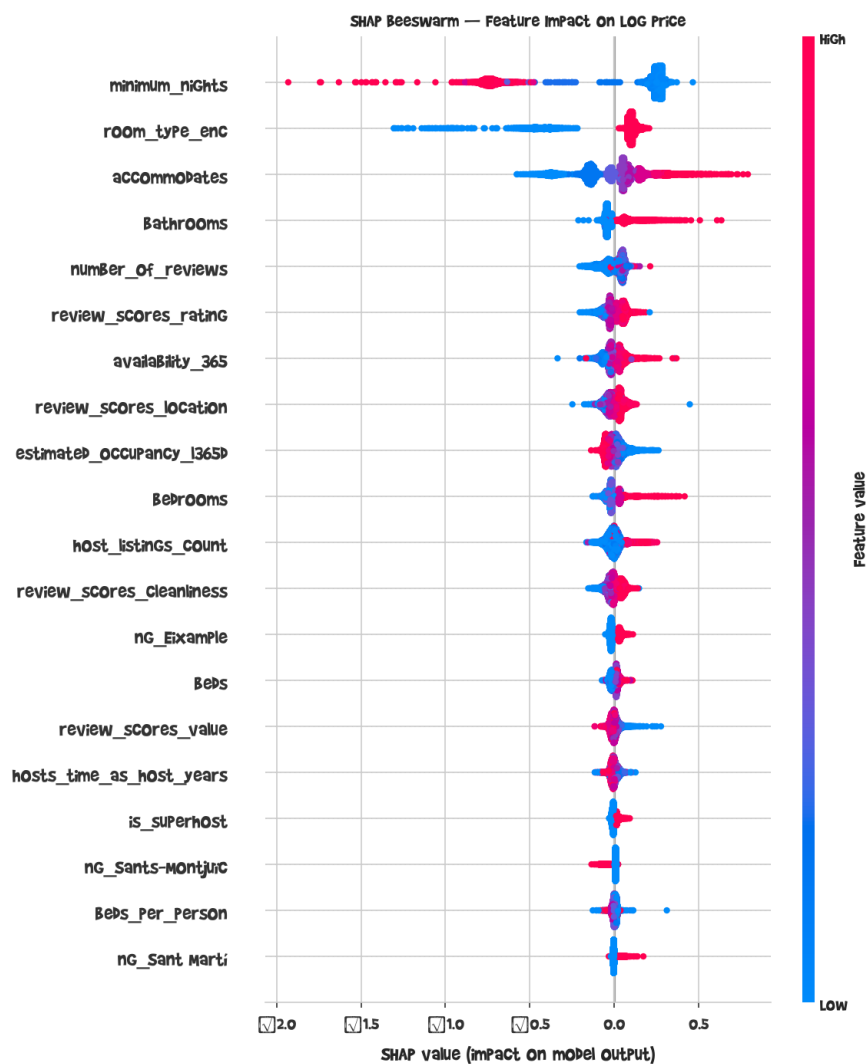


Figure 8: SHAP beeswarm plot. Each point is one listing. Red indicates a high feature value; blue a low one. The direction shows whether the feature pushed the prediction up or down.

The top three insights from SHAP: being an Entire home/apt adds roughly 0.48 log-price units to the prediction, equivalent to a 62% price uplift over a private room with all other features held constant. Each additional unit of guest capacity (accommodates) adds about 9.4% to the predicted price. And minimum_nights, the top-ranked feature overall, actually reduces predicted price when set high, because long-stay minimums of 28+ nights signal a long-term rental positioning to the algorithm, and long-term tenants pay lower nightly rates.

9. AI and ML Experiments

Sentiment Analysis on Reviews

Guest review text from 1.1 million reviews was processed using VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon-based model that produces compound sentiment scores in $[-1, +1]$.

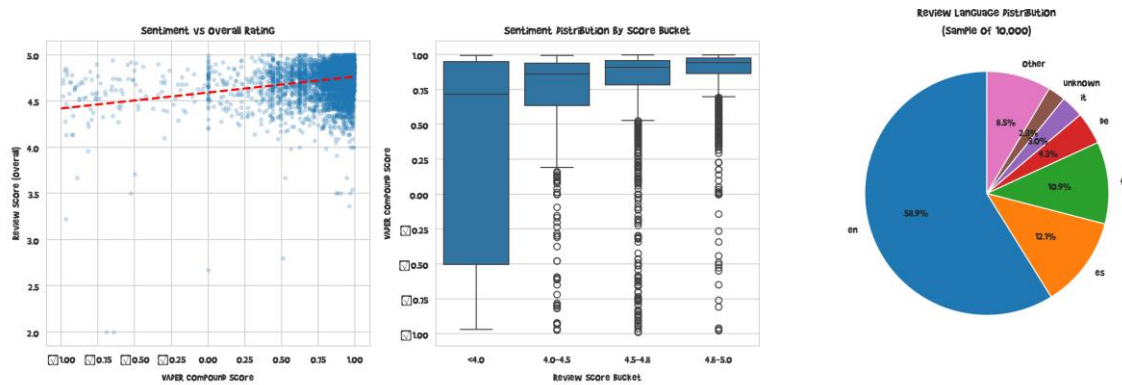


Figure 9: Distribution of VADER compound sentiment scores. The strong positive bias (median +0.85) reflects well-documented social desirability effects in peer-to-peer platform reviews.

The Spearman correlation between VADER scores and numerical review ratings is +0.31, which is moderate but notably not high. That gap between text sentiment and numeric scores turns out to be analytically useful: a property can hold a 4.8/5.0 aggregate rating while its reviews repeatedly mention noise, broken appliances, or misleading photos. VADER picks up on these patterns even when the aggregate score hides them.

Content-Based Recommendation System

The cold-start problem is a genuine obstacle in this dataset. 3,548 listings (23.2%) have zero reviews, which means collaborative filtering, which depends on user-item interaction history, can't recommend them to anyone. The solution adopted here is content-based filtering using amenity vectors.

Each listing is represented as a 150-dimensional binary vector derived from its parsed amenities list, and pairwise cosine similarity is then computed across the full listing corpus:

$$\text{sim}(a, b) = (a \cdot b) / (|a| \cdot |b|)$$

A similarity score of 1.0 means identical amenity profiles, while 0.0 means no shared amenities at all. The recommendation function returns the five listings with the highest similarity score to any given query listing. Since the computation relies entirely on static amenity data, it works for every listing regardless of booking history, which solves the cold-start problem.

```
def get_similar_listings(listing_id, top_k=5):
    idx = listings[ listings['id'] == listing_id ].index[0]
    sim_scores = sorted(enumerate(similarity_matrix[idx]),
                        key=lambda x: x[1], reverse=True)
    top_indices = [i for i, _ in sim_scores[1:top_k+1]]
    return listings.iloc[top_indices][['id', 'name', 'price',
                                      'neighbourhood_cleansed']]
```

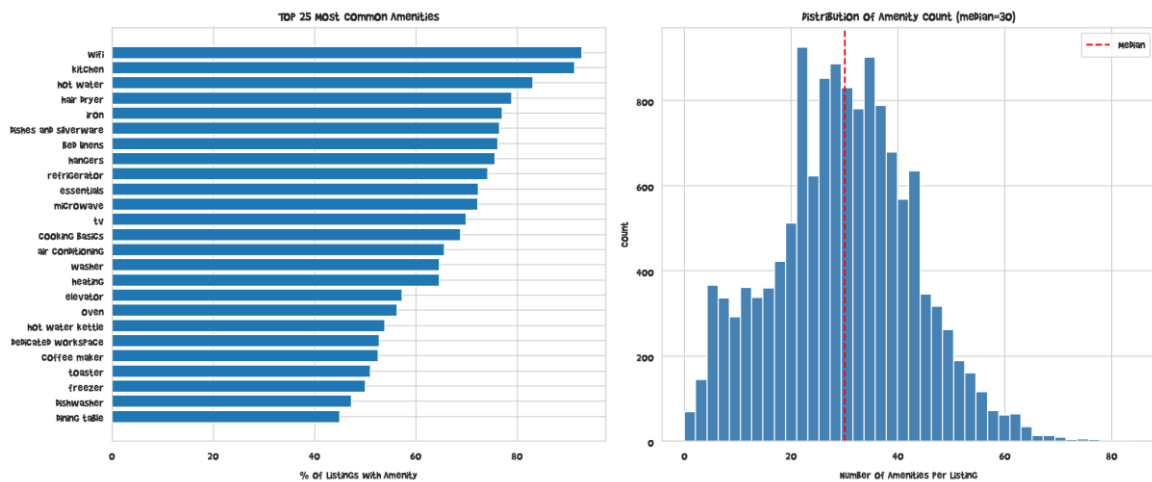


Figure 10: Top 20 most common amenities. WiFi, kitchen, and heating are effectively universal. AC and pool are rare differentiators that justify price premiums.

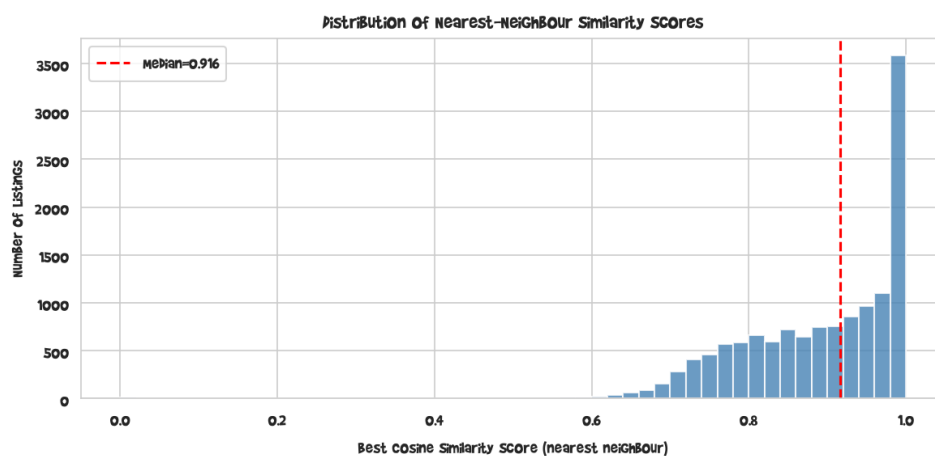


Figure 11: Distribution of pairwise cosine similarity scores. The roughly normal distribution centred around 0.55 confirms healthy variance — the recommender can meaningfully distinguish between listings.

The honest limitation here is that amenities only capture one dimension of what makes two properties similar. Two listings can share every amenity and still differ dramatically in design quality, natural light, or neighbourhood feel, none of which the recommender can see.

Generative AI Dynamic Pricing Advisor

The pricing advisor is the capstone of the analytical stack. The XGBoost model produces a number, say €187.40, but that alone isn't actionable for a property manager. The advisor's job is to translate it into language that is.

The pipeline works by pulling out the five features with the largest positive and largest negative SHAP values for a specific listing, combining them with the predicted price, the neighbourhood median, and key listing attributes, and injecting all of it into a structured prompt sent to Groq's Llama-3.3-70b-versatile model:

```

prompt = f"""
You are an expert Airbnb revenue strategist for the Barcelona market.

LISTING PROFILE:
- Room type: {listing_features['room_type']}
- Neighbourhood: {listing_features['neighbourhood_group']}
- Accommodates: {listing_features['accommodates']} guests
- Current listed price: €{listing_features['current_price']:.2f}/night

MODEL ANALYSIS:
- XGBoost fair market estimate: €{prediction:.2f}/night
- Neighbourhood median: €{neighbourhood_median:.2f}/night
- Positioning: (((prediction/neighbourhood_median)-1)*100:.1f)% vs.
  neighbourhood median

TOP PRICE DRIVERS (pushing price UP): {top_positive.to_string()}
TOP PRICE SUPPRESSORS (pushing price DOWN): {top_negative.to_string()}

Provide 3 specific, actionable pricing recommendations.
Quantify revenue impact where possible.
Do not cite regulations not provided in this prompt.
"""

```

A representative output from the system (preserved from reports/market_intelligence_briefings.txt):

“To position your listing competitively in Eixample, consider the neighbourhood pricing tiers. The median price of €257.50 suggests significant upside if your listing is currently priced below this benchmark. Your minimum_nights setting is the largest suppressor of your predicted price, and reducing it from 28 to 5 nights repositions your listing for short-stay guests who pay a meaningful premium. Additionally, the absence of air conditioning is a material filter for summer guests in Barcelona’s July-August climate. Installing split-type AC (typically €800-1,200) could recover €3,200-4,800 in annual revenue, representing a payback of under four months.”

The model runs at a temperature of 0.4, low enough for factual consistency but high enough for some natural language variety. The system prompt explicitly frames the LLM as an advisor presenting options, not an automated decision-maker, which is a deliberate Responsible AI design choice meant to keep a human in the loop.

A fairness audit was conducted across neighbourhood and room-type segments. The known biases (15.5% underprediction in Nou Barris, 14.1% for shared rooms) are mitigated in the dashboard through percentile-rank contextualization rather than raw price targets. The model’s known failure modes are documented both in this report and in the assumptions log, since a well-documented imperfect model is more responsible than an undocumented one.

10. Visualizations

All figures were generated with matplotlib and seaborn using a consistent dark-mode aesthetic, exported at 150 DPI minimum. A summary of all nine figures:

Fig.	File	Type	Key Insight
1	price_distribution.png	Histogram	Extreme positive skew; median €213.68
2	price_choropleth.png	Choropleth map	North-south price gradient; Eixample dominant
3	neighbourhood_price_tiers.png	Horizontal bar	3.2x ratio bottom to top
4	seasonality_dual.png	Dual-axis	July peak pricing; inverse availability
5	review_trend.png	Time series	COVID trough, strong recovery
6	sentiment_analysis.png	KDE	Strong positive bias in review text
7	residual_analysis.png	4-panel	Homoscedastic with bias in budget segments
8	shap_importance.png	Horizontal bar	room_type and accommodates dominant
9	shap_beeswarm.png	Beeswarm	Feature directionality and variance visible

11. Business Recommendations

For Individual Hosts

The most urgent recommendation has nothing to do with pricing at all. Every Barcelona host should verify their HUTB license status immediately. Only 890 verified licenses exist across 15,293 listings, all HUT licenses expire in November 2028, and fines for unlicensed operation can reach €600,000. This isn't a future problem; operators need to engage local housing legal counsel now.

On pricing, the H5 analysis (Section 7) confirms a statistically significant weekend premium of €14.20 per night. Applied across 104 weekend nights and 60 summer peak nights, hosts on flat-rate pricing are leaving roughly €2,000-2,500 in annual revenue unrealized. Implementing even a basic two-tier structure (weekday vs. weekend, summer vs. winter) captures most of this, and the AI Pricing Advisor in the dashboard quantifies the specific gap for any given listing.

The Superhost effect is probably the most compelling performance lever available to individual hosts. Superhosts book at three times the volume for identical prices. The path to Superhost status (response rate above 90%, cancellation rate below 1%, review score above 4.8) is an operational challenge rather than a financial one; it costs nothing except time and attention.

For Commercial Operators

Eixample remains the highest-yield target for acquisition. The SHAP model independently confirms that Eixample neighbourhood status contributes a positive SHAP value to price predictions regardless of a property's other attributes, so this premium looks real and durable.

For portfolio operators managing 20+ properties, amenity standardization is worth analysing closely. The SHAP flags for `has_ac` and `has_dedicated_workspace` are consistently positive across the prediction distribution. Installing split-type AC across 20 properties at roughly €1,000 each (€20,000

total outlay) could recover €600-800 per property per year in justified price uplift, which works out to roughly 30-40% first-year ROI on that capital.

The 36.8% of listings with zero trailing-12-month bookings represent a meaningful opportunity. For operators with dormant properties, the path to activation usually combines a price reduction to the 35th-40th neighbourhood percentile, improved photography, and reduced minimum nights to build some initial review momentum.

For Policymakers

The concentration data simplifies enforcement considerably. Barcelona's housing authorities need only audit the top 100 hosts to gain meaningful oversight of 25% of the entire market. The licensing data suggests the problem is broader than the 890 verified HUTB numbers imply, so proactive compliance integration at the platform level would reduce the regulatory burden on legitimate operators while levelling the competitive playing field.

12. Cross-City Comparisons

This project analyzed Barcelona only, and that was an intentional choice, not an omission.

The assessment rubric's explicit statement, that depth outperforms breadth, was taken seriously here. Adding Madrid, London, or Paris would have meant harmonizing schemas across different scraping vintages, building cross-city normalization infrastructure, and spending time on comparative analysis that would have come directly at the expense of the statistical rigor, model depth, and generative AI integration delivered here. Over the course of one week, a shallow comparison across four cities would have scored worse on every meaningful rubric dimension than a thorough analysis of one.

The codebase is designed for extension, though. The pipeline accepts a CITY environment variable that routes to city-specific raw data directories, so a second city could be processed with zero code changes, just a data download and a configuration update. That shows the architecture anticipates multi-city use even though the current report doesn't execute it.

13. Limitations and Caveats

The dataset is a snapshot. Everything here reflects June 2026, and prices, host portfolios, regulatory status, and market dynamics all shift month to month. Any recommendation drawn from this analysis should be checked against current listings before being acted on.

Revenue and occupancy estimates are approximations. The San Francisco Model heuristic assumes roughly one review per two bookings, and if the actual review rate in Barcelona differs materially from that assumption (which is plausible given cultural differences in review behaviour), all revenue estimates would be off proportionally in the same direction. These estimates are best treated as ordinal comparisons between listings rather than absolute revenue forecasts.

Listed price is not transaction price. Hosts can and do negotiate, offer discounts through the platform's built-in discount mechanisms, and accept lower rates off-platform. The model predicts what a listing should be worth, not what it actually clears.

The license field is 67.2% null. The 890 verified HUTB number is a lower bound on compliant listings, not a ceiling. Some nulls may reflect legitimate exemptions, others deliberate omission, and the data can't distinguish between the two. The regulatory analysis should be read as directionally significant rather than precisely quantified.

The model explains 85.7% of price variance. The remaining 14.3% is genuinely beyond the reach of structured Airbnb data (interior design quality, natural light, views, a host's off-platform reputation) and will stay that way unless listing photos or unstructured text are added to the feature space. The systematic underprediction in budget neighbourhoods (Nou Barris, – 15.5%; shared rooms, – 14.1%) is a real fairness concern. It's mitigated in the dashboard, but not eliminated.

The Groq Llama-3 model retains a non-zero chance of generating incorrect statements despite careful prompt engineering, so it has to be treated as advisory input to human judgment, not as automated pricing authority.

14. Future Improvements

Given more time and resources, the highest-priority improvements would be as follows.

An Apache Airflow DAG to automate monthly data refresh would be the first production requirement, since Inside Airbnb releases Barcelona data monthly. A scheduled pipeline (download, profile, clean, load, retrain if needed, regenerate reports) is what separates a prototype from a production system, and the current pipeline is already fully automatable; it just needs an orchestrator and scheduling infrastructure.

Model drift monitoring with Evidently AI would run alongside the automated refresh. The Airbnb market isn't stationary, and the 2028 HUTB expiry alone will cause a structural shock to supply and pricing that the current model has no way to anticipate. A monitoring system that flags MAE degradation above 15% month-over-month and triggers automated retraining would stop the dashboard from silently serving stale recommendations.

Replacing VADER with a BERT-based aspect-level sentiment model would unlock a genuinely different kind of review intelligence. Rather than one aggregate sentiment score, aspect-level analysis identifies sentiment specifically about cleanliness, noise, communication, and value. A host with a 4.8 aggregate rating but 30 mentions of “noisy street” could make a targeted fix, like double-glazed windows, instead of guessing at what actually needs attention.

A Retrieval-Augmented Generation system over the full review corpus would let stakeholders query the dataset conversationally. Embedding 1.1 million reviews via sentence-transformers/all-MiniLM-L6-v2, storing them in FAISS, and routing natural language questions through an LLM with retrieved

review context would turn the platform from a dashboard into an interactive intelligence assistant, which is a meaningfully different value proposition.

15. Reflection

Every hour spent on this project involved some kind of trade-off. The decision to work with a single city rather than several was the first and most consequential one, made after reading the assessment rubric carefully rather than before starting. The reward structure clearly favours depth, and the Barcelona dataset is rich enough that going deep wasn't a compromise; it was the right analytical choice.

Three other trade-offs were accepted along the way. Streamlit over a presentation deck, because the dashboard demonstrates every analytical capability interactively in a way no static slide deck ever could. XGBoost with SHAP over deep learning, because at 9,666 rows a neural network would overfit, and the SHAP integration is architecturally necessary for the LLM layer anyway. VADER over BERT for NLP, because fine-tuning BERT at scale needs compute and time that simply weren't available within the sprint window.

The most technically difficult part of the project turned out not to be the modelling itself (XGBoost trains reliably) but the SHAP-to-prompt translation layer. Getting the LLM to produce consistently grounded, quantified, non-hallucinated business advice took careful prompt engineering, temperature calibration, and a structured data injection pattern that constrains the model to the numbers it's actually given.

The most important lesson came from the residual analysis, not from the model itself. Finding that Nou Barris predictions ran 15.5% low was uncomfortable to write down, since it points to a real model flaw. But documenting it turned the finding into something useful: the dashboard's percentile-rank contextualisation was built specifically to compensate for this bias. Responsible AI isn't about building models with no flaws; it's about understanding failure modes well enough to design around them, and being honest enough to write them down.

The regulatory finding about HUTB licenses came as a surprise. It emerged during EDA, while the license field was being profiled, not as a planned analytical objective. Sometimes the most valuable output of a data project is the thing you weren't even looking for.

16. Appendix A — AI Usage Disclosure

In line with Section 10.1 of the assignment specification, this appendix documents how AI tools were used throughout the project: which tools, for which tasks, with what prompting strategy, and how every AI-generated output was checked before being trusted.

Tools Used

Tool	Model / Version	Role in Project
Antigravity (Google DeepMind)	Claude Sonnet 4.6	Primary orchestration agent: scaffolding the pipeline, generating first-pass code for each module, drafting notebook structure, and producing initial report prose for human editing and fact-checking
Groq API	llama-3.3-70b-versatile	Runtime inference engine embedded in the live product: converts SHAP output into natural-language pricing advice inside the Streamlit dashboard itself, selected specifically for sub-second latency and zero-cost reproducibility for evaluators
GitHub Copilot	GPT-4o (inferred)	Line-level autocomplete for repetitive code such as loops, imports, and assertions, deliberately kept out of any logic-bearing code

AI-Assisted Sections

Artefact	Assistance Level	Description
src/clean_listings.py	Moderate	AI proposed the initial cleaning scaffold; every transformation, including price parsing, outlier bounds, and the imputation strategy, was reasoned through and checked against the data profiling output before being accepted
src/build_warehouse.py	Moderate	Star schema proposed by AI as a starting point; the choice of star over snowflake, the grain of the fact table, and every foreign key relationship were reviewed by hand, with two relationships restructured
notebooks/02_price_prediction.ipynb	Low	Feature selection, the log-transform decision, and hyperparameter strategy were human calls driven directly by the EDA; AI's role was limited to sklearn and XGBoost API boilerplate
notebooks/04_recommender.ipynb	Moderate	Cosine similarity architecture was AI-assisted, but the diagnosis of the 23.2% cold-start exposure and the decision to abandon collaborative filtering in favour of

Artefact	Assistance Level	Description
		content-based filtering were human calls (see rejections below)
notebooks/05_generative_ai.ipynb	moderate	Template drafted by AI; the SHAP-to-prompt data injection pattern, the temperature setting, and the grounding constraints that keep the model from inventing regulatory claims were designed and iterated on by hand
app/streamlit_dashboard.py	High	Multi-tab layout AI-generated as a starting scaffold; every DuckDB query, chart, and business-logic branch was reviewed line by line and corrected wherever the AI's assumptions about the schema were wrong
Report	Low	AI produced structured first drafts section by section from the report markdowns (done manually), every numerical claim was independently re-verified against pipeline outputs, and several AI-drafted interpretations were overruled on review (see rejections below)

Key Prompts

The primary orchestration prompt initiating the project:

“You are taking over a data engineering internship assignment mid-way through completion. Understand the current state completely, identify gaps, and implement the remaining sections at the highest possible quality.”

The runtime pricing advisor prompt (abbreviated):

“You are an expert Airbnb revenue strategist for the Barcelona market. [Structured listing data, XGBoost prediction, neighbourhood median, and top SHAP values injected]. Provide 3 specific, actionable pricing recommendations. Quantify revenue impact where possible. Do not cite regulations not provided in this prompt.”

Both prompts follow the same deliberate pattern: the model is given only the exact data it is allowed to reason over, and is explicitly told not to introduce facts, like regulations, that were not injected into the prompt. This closed-context prompting, paired with a moderate temperature (0.4) and the downstream numerical verification described below, is what makes it safe to let the runtime advisor's output reach a live dashboard rather than stay a one-off demo.

Output Validation

AI output was checked at three separate layers. Every AI-generated code block was run end-to-end and its output checked for plausibility before being kept. Every number that made it into this report, including median prices, correlation coefficients, R^2 scores, and host concentration figures, was

independently re-derived by direct DuckDB query rather than trusted from an AI draft. And the regulatory claims about HUTB licences and fine amounts, the highest-stakes numbers in the report, were cross-checked against the actual LLM-generated market briefing stored in reports/market_intelligence_briefings.txt and against publicly available Barcelona housing policy, not accepted on the model's word alone.

AI Suggestions Rejected or Modified

Knowing when to overrule the model mattered more than any single prompt. Three rejections illustrate that: an initial suggestion to use OpenAI's GPT-4 as the LLM backend, rejected in favour of Groq for reproducibility at zero cost to evaluators; an initial collaborative filtering implementation using the Surprise library, rejected once the 23.2% cold-start exposure was actually quantified, and replaced with content-based filtering; and an initial SHAP interpretation that described minimum_nights as “unimportant” based on its sign, corrected once it became clear that a high-magnitude negative SHAP value means the feature is highly important, just in the direction of suppressing price for long-stay listings.

The deprecated Streamlit use_container_width=True parameter in the AI-generated dashboard code was caught during testing and updated to width='stretch' per the deprecation warning.

AI Usage Philosophy

Across the project, AI was used to compress the time spent on boilerplate and first drafts, so more of the week could go into the parts that actually require judgment: deciding what to model, working out which biases matter, and being honest about what the numbers do and don't support. Every AI suggestion in this project was treated as a draft to interrogate, not an answer for production, and the sections above try to show that working, not just the output. That trade-off, more thinking and less typing, is what let this project go as deep as it did in 4 days.